AttentionDCA on the J-domain family: pseudo-likelihood training, sampling and generative checks

TP4 — remastered

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Abstract

We fit a Potts model to the bacterial J-domain alignment ($L = 63$ sites, $q = 21$ states) with couplings parametrised by a factored self-attention layer (AttentionDCA), trained by maximising the reweighted pseudo-likelihood. New sequences are drawn with the pseudo-likelihood (Gibbs-like) conditional sampler, and compared with a held-out natural set through PCA, one- and two-site statistics, Hamming distances and the energy trajectory of the chains. The supplied train/test files turned out to share 79% of their sequences verbatim, so the split is rebuilt from the full 36 713-sequence pool by clustering at 80% identity; all results below use that sequence-disjoint split. With $H = 128$ heads and $d = 10$, sequences generated at $\beta = 1$ reproduce the natural site frequencies ($r = 0.990$) and connected correlations ($r = 0.871$, against a train-vs-test ceiling of 0.944), while being new (mean distance 0.42 to the closest training sequence) and as diverse as natural sequences (mean pairwise distance 0.68 vs. 0.60). Chains forget their starting sequence in about 11 sweeps ($\tau_{\text{int}} = 6.5$ sweeps for the energy). The one clear defect is a minor HscB-like sub-family holding 14.1% of the natural weight but only 5.1% of the samples; restricting the comparison to the modes the sampler does populate raises the correlation agreement to 0.918, near the 0.941 ceiling.

1 Model and theory

Potts model / DCA. An aligned sequence $\boldsymbol{x} = (x_1, \dots, x_L)$, $x_i \in \{1, \dots, q\}$ (20 amino acids + gap), is modelled as

$$P(\boldsymbol{x}) = \frac{1}{Z}\, e^{-E(\boldsymbol{x})}, \qquad E(\boldsymbol{x}) = -\sum_{i<j} J_{ij}(x_i, x_j) - \sum_i h_i(x_i). \tag{1}$$

The couplings $J_{ij}(a, b)$ capture co-evolution between positions (Direct Coupling Analysis). As in the original AttentionDCA, no explicit fields are used ($h_i \equiv 0$): conservation is absorbed in the couplings.

Attention parametrisation. With H heads, each with query/key matrices $Q^h, K^h \in \mathbb{R}^{d \times L}$ and a value matrix $V^h \in \mathbb{R}^{q \times q}$,

$$A^h_{ij} = \frac{\exp\big(\sum_k Q^h_{ki} K^h_{kj}\big)}{\sum_{j' \neq i} \exp\big(\sum_k Q^h_{ki} K^h_{kj'}\big)} \;(A^h_{ii} = 0), \qquad J_{ij}(a, b) = \sum_{h=1}^{H} A^h_{ij}\, V^h(a, b). \tag{2}$$

Positional structure (which sites interact, A^h) is separated from amino-acid chemistry (V^h); the model has $H(2dL + q^2)$ parameters instead of $\binom{L}{2}q^2$. Because A^h is row-normalised, $J_{ij}(a, b) \neq J_{ji}(b, a)$ in general.

Pseudo-likelihood. The partition function is intractable, so we maximise the (weighted) pseudo-likelihood built from the single-site conditionals

$$p(x_i = a \mid \boldsymbol{x}_{\setminus i}) = \frac{\exp\big(\sum_{j \neq i} J_{ij}(a, x_j)\big)}{\sum_c \exp\big(\sum_{j \neq i} J_{ij}(c, x_j)\big)}, \qquad \mathcal{L} = -\sum_{m=1}^{M} w_m \sum_{i=1}^{L} \log p\big(x^m_i \mid \boldsymbol{x}^m_{\setminus i}\big) + \lambda \sum_{i \neq j} \sum_{a,b} J_{ij}(a, b)^2, \tag{3}$$

with $\sum_m w_m = 1$. Training uses Adam (lr $5 \cdot 10^{-3}$, batches of 1000, per-batch renormalised weights), a 10% validation split and early stopping on the validation pseudo-likelihood.

Sequence reweighting. Natural alignments over-represent some clades. Each sequence gets

$$w_m = \frac{1}{\big|\{n :\; d_H(\boldsymbol{x}^m, \boldsymbol{x}^n) < \theta L\}\big|}, \qquad M_{\text{eff}} = \sum_m w_m, \qquad \theta = \min\Big(0.5,\; \frac{0.38 \cdot 0.32}{\langle \text{fraction identity} \rangle}\Big), \tag{4}$$

(d_H = Hamming distance, the sequence itself included). The same weights are used for the statistics of the natural sets in §2; this gives $\theta = 0.312$, $M_{\text{eff}} = 3590$ for the 29 312 training sequences.

Pseudo-likelihood sampling. Sequences are generated with the trained conditionals: in each *sweep* the L sites are visited in random order and x_i is resampled from $p_\beta(x_i = a \mid \boldsymbol{x}_{\setminus i}) \propto \exp\big(\beta \sum_{j\neq i} J_{ij}(a, x_j)\big)$. For symmetric couplings this is exact Gibbs sampling of $P(\boldsymbol{x})^\beta$; with the asymmetric attention couplings it is a "pseudo-Gibbs" chain whose stationary law is only approximately Boltzmann. We therefore monitor two scores along the chains:

$$E(\boldsymbol{x}) = -\tfrac{1}{2}\sum_{i\neq j} J_{ij}(x_i, x_j) \quad \text{(symmetrised Hamiltonian)}, \qquad \log \mathrm{PL}(\boldsymbol{x}) = \sum_i \log p(x_i \mid \boldsymbol{x}_{\setminus i}). \tag{5}$$

Decorrelation is measured by the chain-averaged normalised autocorrelation $\rho(t) = \langle \delta E_s \, \delta E_{s+t}\rangle / \langle \delta E^2 \rangle$ at stationarity, the integrated autocorrelation time $\tau_{\mathrm{int}} = 1 + 2\sum_{t\geq 1}\rho(t)$ (Sokal window), and the normalised Hamming distance to the starting sequence $d_H(\boldsymbol{x}_t, \boldsymbol{x}_0)/L$.

2 Evaluation metrics

Site frequencies and correlations. For a set with weights w_m ($\sum w_m = 1$; uniform for generated sets)

$$f_i(a) = \sum_m w_m \, \delta_{x_i^m, a}, \quad f_{ij}(a,b) = \sum_m w_m \, \delta_{x_i^m,a}\delta_{x_j^m,b}, \quad C_{ij}(a,b) = f_{ij}(a,b) - f_i(a) f_j(b) \ (i<j). \tag{6}$$

Generated and natural statistics are compared entry-wise ($Lq = 1\,323$ values of f_i, $\binom{L}{2}q^2 = 861\,273$ values of C_{ij}) with the Pearson coefficient $r = \mathrm{cov}(u,v)/(\sigma_u\sigma_v)$, the Spearman rank coefficient, the least-squares slope of generated vs. natural (1 = correlations reproduced at the right amplitude) and RMSE $= \sqrt{\langle (u-v)^2\rangle}$. Two references bracket the numbers: *train vs. test* (finite-sample ceiling) and an *independent-site model* sampled from the training f_i (reproduces f_i, has $C_{ij} \equiv 0$).

PCA. Sequences are one-hot encoded ($\boldsymbol{x} \mapsto \{0,1\}^{Lq}$); the two leading principal components are fitted on the natural test set only, and every other set is projected onto them.

Hamming distances. $d(\boldsymbol{x},\boldsymbol{y}) = \frac{1}{L}\sum_i (1 - \delta_{x_i,y_i})$. Novelty: $d_{\mathrm{nn}}(\boldsymbol{x}, S) = \min_{\boldsymbol{y}\in S} d(\boldsymbol{x},\boldsymbol{y})$ to the training/test set. Diversity: distribution of d over all generated–generated pairs, compared with test–test pairs.

3 Data: audit of the supplied files and a sequence-disjoint split

What the supplied files contain. The project ships several bacterial J-domain alignments ($L = 63$ match columns). Table 1 lists them with the fraction of their unique sequences that also occur in `jdoms_bacteria_filtered.fasta` and in `jdoms_bacteria_train2.fasta`. Three facts come out of it:

- `jdoms_bacteria_filtered.fasta` (36 713 unique sequences) is a *superset*: every unique sequence of every other bacterial file is contained in it.
- `filtered2` is exactly `train2` ∪ `test2` (100% of each is inside it), so that pair is a partition of a single pool.
- `train2` and `test2` have disjoint UniProt identifiers, but 78.8% of the `test2` sequences are *byte-identical* to a `train2` sequence (mean distance to the nearest training sequence 0.044). The split was made by species, and the same J-domain sequence recurs in many species, so it does not hold out sequences at all. No pair of supplied files is clean: the best is `train`→`test2` at 17% exact copies, but that training file has only 7 695 unique sequences.

The split used here. We therefore rebuild the split from the 36 713-sequence superset. Sequences are clustered greedily (CD-HIT style: visit from least to most gapped, each unassigned sequence becomes a representative and absorbs everything within 0.2 normalised Hamming distance, i.e. 80% identity), which gives 10 942 clusters (6 767 singletons). Whole clusters are then assigned to train or to test, so that no cluster is split across the two sides, and the only sequences removed afterwards are those *identical* to a training sequence (0 here, since whole clusters move together). The result is 29 312 training and 7 401 test sequences with **no exact copies**: median test-to-nearest-train distance 0.175, mean 0.179, 5% quantile 0.032 and minimum 0.016 (1 substitution). Weights: $\theta = 0.312$, $M_{\mathrm{eff}} = 3590$ (train) and $\theta = 0.303$, $M_{\mathrm{eff}} = 1280$ (test).

A test sequence that merely *resembles* a training sequence is deliberately kept: greedy clustering guarantees that cluster representatives are 0.2 apart but not that every pair of members of different clusters is, so 61%

file	sequences	unique	% in `filtered`	% in `train2`
`filtered`	36 713	36 713	100%	40%
`filtered2`	18 241	18 241	100%	80%
`train`	9 999	7 695	100%	33%
`test`	50 125	31 670	100%	39%
`train2`	14 502	14 502	100%	100%
`test2`	3 739	3 739	100%	79%
`test_4000`	3 999	3 384	100%	33%

Table 1: The supplied bacterial J-domain files. The last two columns are the fraction of a file's *unique* sequences that also appear in `filtered` and in `train2`.

of the test set still lies within 0.2 of some training sequence. Discarding those would leave only the sparse, isolated corners of the family and would misrepresent its density — precisely the density the generative model is asked to reproduce. What the split guarantees is therefore the thing that matters for a likelihood: no test sequence was ever seen during training. Section 7 compares the two splits.

4 Training

A note on units: nats. All log-likelihoods here use the natural logarithm, so they are measured in *nats* — the information-theoretic unit you get from ln instead of $\log_2$ (1 nat $= 1/\ln 2 \approx 1.44$ bits). A model that assigns probability p to a sequence scores $-\ln p$ nats on it, so a difference of Δ nats between two models means one assigns the data e^{Δ} times more probability: 0.7 nats is a factor of two, 2.3 nats a factor of ten. Because $\log \mathrm{PL}$ sums $L = 63$ site terms, a value of -78.72 nats per sequence is about -1.25 nats per site, against $\ln 21 = 3.04$ for uniform guessing.

Hyper-parameters. Table 2 is a grid over H, d and λ, scored by the weighted $-\log \mathrm{PL}$ on a 10% validation split of the training file (maximum 3000 epochs, patience 50, best epoch restored). Doubling H helps until about 128 heads, after which the gain is 0.3 nats per sequence for twice the parameters. We keep $H = 128$, $d = 10$, $\lambda = 10^{-3}$ (217 728 parameters, best epoch 668). The training curve is in Fig. 1. Weighted $-\log \mathrm{PL}$ per sequence: 71.10 (train) and 78.72 (test). The test set here shares no sequence with the training set, so that gap (7.6 nats per sequence, about 11%) is a genuine generalisation gap rather than memorisation.

Structure check. The couplings carry structural information: after symmetrisation, zero-sum gauge, Frobenius norm and APC, the top $L/2$ and top L ranked pairs with $|i - j| > 4$ are contacts (< 8 Å in the J-domain structure) with precision 1.00 and 0.83 (random pairs: 0.21).

H	d	λ	params	best epoch	val. $-\log \mathrm{PL}$
256	10	10^{-3}	435 456	187	85.33
128	10	10^{-3}	217 728	387	85.61
64	10	10^{-3}	108 864	709	86.73
64	10	10^{-4}	108 864	168	87.34
64	23	10^{-4}	213 696	293	87.37
64	23	10^{-3}	213 696	654	87.63
32	10	10^{-4}	54 432	817	87.78
32	23	10^{-4}	106 848	697	88.17
32	23	10^{-3}	106 848	2061	88.22
32	10	10^{-3}	54 432	817	89.10
16	10	10^{-4}	27 216	973	89.74
16	23	10^{-4}	53 424	820	90.61
16	23	10^{-3}	53 424	2024	90.64
16	10	10^{-3}	27 216	1692	91.34

Table 2: Hyper-parameter grid, sorted by validation $-\log \mathrm{PL}$ per sequence (weighted, no penalty). Chosen model: second row.

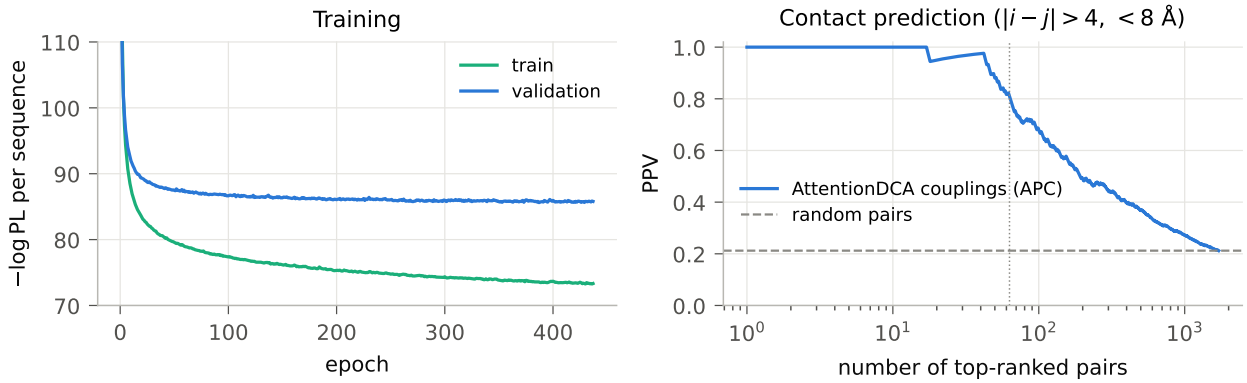


Figure 1: Left: weighted $-\log \mathrm{PL}$ per sequence (training batches vs. validation split). Right: contact-prediction precision of the ranked APC-corrected coupling norms; the vertical line marks L pairs.

5 Sampling and decorrelation

Implementation. All chains are updated at once on the GPU: the conditional logits of site i for C chains are a single $(C \times Lq) \cdot (Lq \times q)$ product. One site-update per chain costs about 0.3 μs, against 0.6 ms (NumPy) to 28 ms (torch tensor) in the original pure-Python loop, a speed-up of 10^3–10^5. The original code also saved the state after *every single-site update* as a new sample, so consecutive "sequences" differed at ≤ 1 site. Here every sample is the final state of an independent chain.

Temperature. Pseudo-likelihood fits the conditionals, not the global normalisation, so the $\beta = 1$ chain need not have the energy of natural sequences. Figure 3 scans β (3000 chains, 1000 sweeps). At $\beta = 1$ the chains sit slightly above the natural energy, and increasing β trades diversity for sharper correlations. We report $\beta = 1$ (the learned model as is) and $\beta^* = 1.05$, the value at which the mean energy of the samples equals the reweighted mean energy of the *training* set (-134.3). β^* is therefore chosen without looking at the test set.

Decorrelation from the initial sequence. Figure 2 follows 1000 chains started from training sequences and 1000 from uniformly random sequences over 2000 sweeps.

- Random starts ($E \approx -0.0$) fall into the family basin within about 20 sweeps and reach the same stationary energy as natural starts (-126.6 vs. -126.6 at $\beta = 1$). The stationary state therefore does not depend on the start.
- Chains started from natural sequences first rise in energy (natural sequences are more stable than typical samples at $\beta = 1$), then fluctuate around the stationary mean.
- The distance from the start reaches 90% of its plateau after 11 sweeps. The plateau (0.66) is at or above the mean test–test distance (0.60): the chain forgets its seed entirely.
- Stationary energy autocorrelation: $\tau_{\mathrm{int}} = 6.5$ sweeps at $\beta = 1$ and 5.6 at β^* (PLL: 3.8 and 3.4). Generation uses 500 sweeps per chain ($\approx 40\,\tau_{\mathrm{int}}$), 10 000 independent chains per temperature.
- No chain ends in a degenerate gap-only state (0 of 10 000; $> 90\%$ gaps is the filter applied to the natural MSA). Such a collapse *did* occur (up to 40% of chains) with an under-trained model (150 epochs, $H = 32$), where the all-gap sequence was the energy minimum. It disappears once training converges.

6 Generated vs. natural sequences

PCA (Fig. 4). The first two components of the one-hot test set explain 11.3% and 4.3% of the variance. k-means with $k = 3$ on the (PC1, PC2) plane of the natural test set splits the family into two large clusters and one minor one (the elbow of the within-cluster inertia sits at $k = 3$). Writing s_k for a cluster's share of a set's total weight, a cluster counts as *covered* when $s_k^{\mathrm{gen}}/s_k^{\mathrm{nat}} \geq 0.6$ for both temperatures. 2 of the 3 clusters are covered, holding 85.9% of the natural test weight; the remaining minor cluster (HscB-like J-domains) holds 14.1% of the natural test weight but only 5.1% ($\beta = 1$) and 3.2% (β^*) of the samples. About one chain in ten *starts* inside it, so this is not a failure to reach the cluster: the model gives it too little stationary weight. Over the covered region the agreement is good — the 1-D Wasserstein distances to the reweighted test distribution along PC1/PC2 are 0.26/0.34 ($\beta = 1$) and 0.30/0.31 (β^*), against 1.05/0.66 for the independent-site model.

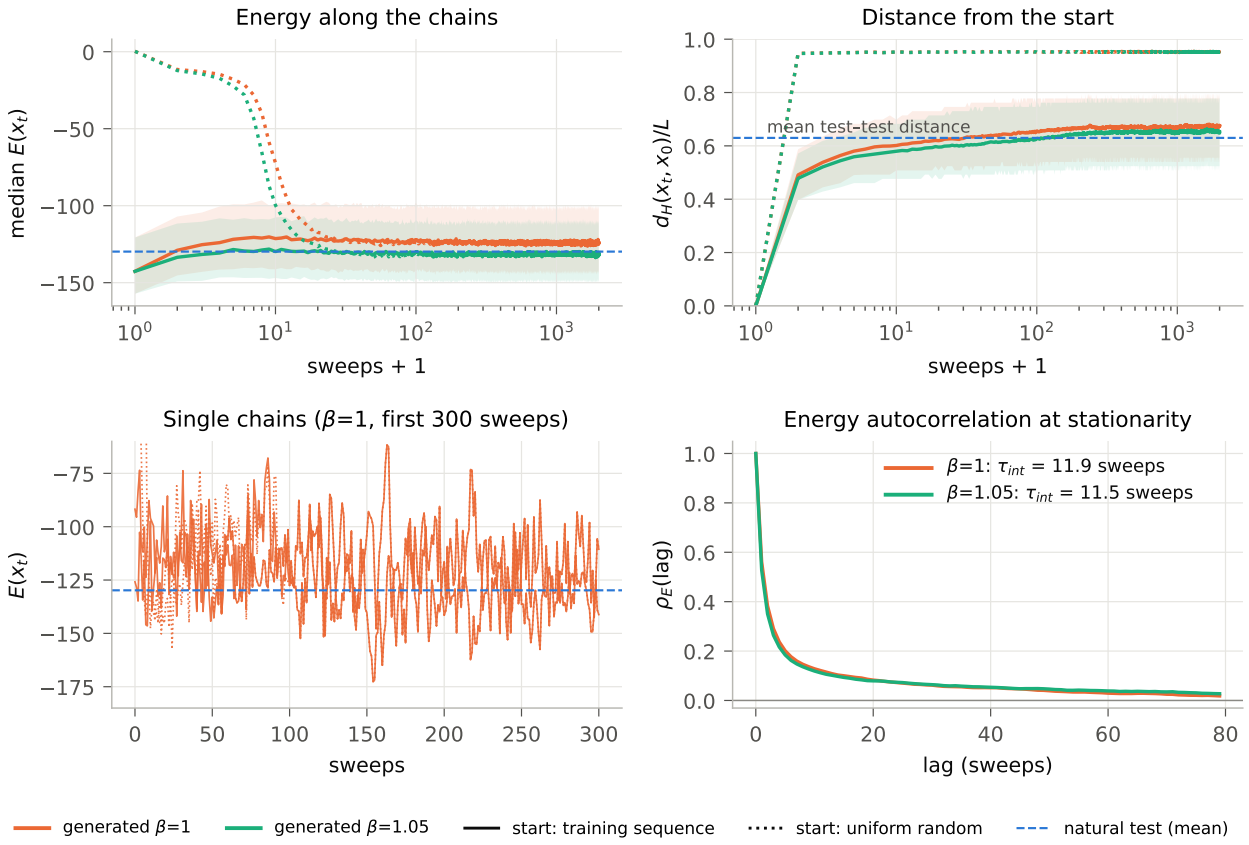


Figure 2: Pseudo-likelihood sampling. (a) median energy (band: 10–90% of chains from training starts); (b) normalised Hamming distance to the starting sequence; (c) individual chains at $\beta = 1$ (solid: training start, dotted: random start); (d) chain-averaged energy autocorrelation over the second half of the chains.

Why the minor cluster is missed: the model does not learn the priors. Table 3 separates the two things that set a cluster's share of the samples. The pseudo-likelihood of *training* sequences is essentially the same in all three clusters (-71.8 nats for C0 against -71.1/-71.0 for the covered ones), so the model has learned the local statistics of the minor sub-family about as well as everything else. Its *energy*, however, is not comparable at all: natural sequences of C0 score -119.5 on the test set and -122.6 on the training set, roughly 14 units above the covered clusters (-134.0/-133.5 and -134.5/-137.2). Since the stationary weight of a basin is set by its free energy, that offset is what starves the cluster: the measured free-energy error is $\Delta F = -\ln(s^{\mathrm{gen}}/s^{\mathrm{nat}}) = +1.01$ nats for C0, i.e. the sampler gives it 0.36 of its natural weight, against $|\Delta F| \leq 0.23$ for the other two.

This is the expected failure mode of this objective. Maximising the pseudo-likelihood constrains the conditionals $p(x_i \mid \boldsymbol{x}_{\setminus i})$ — purely local quantities — and never the global normalisation, so nothing in the loss asks the model to reproduce how *much* total probability mass each sub-family carries. The model has no fields and no mixture component to hold such a prior; the relative weight of the basins is only an indirect consequence of the couplings. Reweighting compounds this, since it is designed to flatten exactly the clade-frequency signal that would tell the model how common each sub-family is. The result is a model that describes every sub-family correctly and populates them in the wrong proportion. It is also why the independent-site baseline distributes its samples as $(0.02, 0.10, 0.88)$ against the natural $(0.14, 0.39, 0.47)$: cluster identity is a covariation effect, and a model without couplings collapses onto the single dominant mode.

Statistics restricted to the covered clusters (Fig. 5, Table 4). To separate "the model distorts the covariation" from "the model misses one sub-family", the whole comparison is repeated with *both* sets restricted to the covered clusters (natural: 6 761 sequences, $M_{\mathrm{eff}} = 1102$; weights recomputed inside the subset). The correlation agreement rises from 0.871 to 0.918 at $\beta = 1$ and from 0.859 to 0.918 at β^*, while the train-vs-test ceiling barely moves (0.944 → 0.941): inside the modes it populates, the model is close to the finite-sampling limit, and the missed sub-family accounts for most of the residual gap. On the strongest 1% of correlations the restricted agreement is 0.961 / 0.961, with the β^* slope at 0.91.

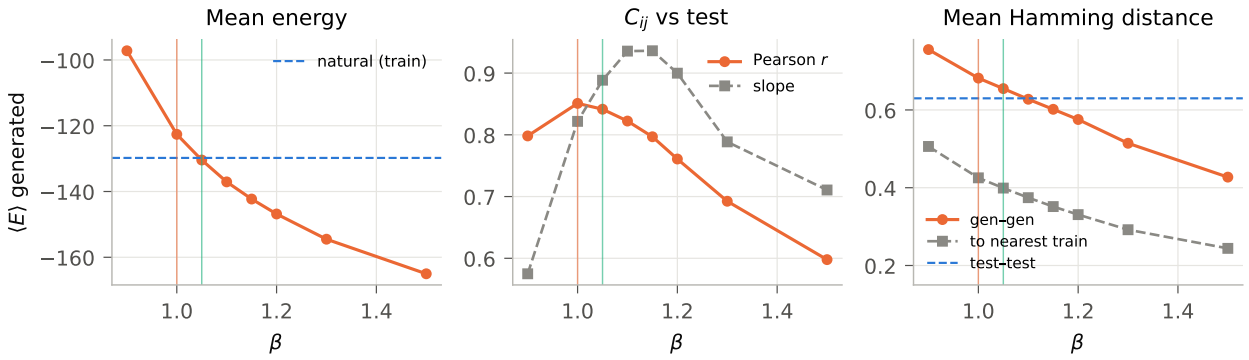


Figure 3: Inverse-temperature scan (chains started from training sequences). Vertical lines mark $\beta = 1$ and $\beta^* = 1.05$. Right: mean pairwise distance among samples (gen–gen), mean distance to the closest training sequence, and the natural test–test value.

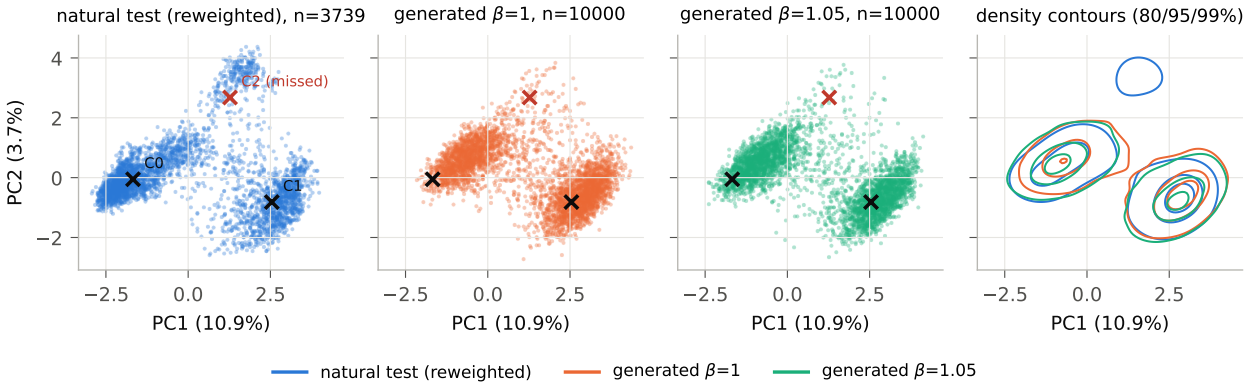


Figure 4: Test and generated sequences projected on the first two principal components of the one-hot test alignment (PCA fitted on the test set only); each panel draws a random sample of the points, the natural panel drawn with its reweighting weights. Crosses: k-means centres, red for the cluster the samples miss. Right: densities of the full sets.

One- and two-site statistics (Fig. 6, Table 5). Site frequencies are reproduced almost exactly ($r = 0.990$). The connected correlations reach $r = 0.871$ at $\beta = 1$ and $r = 0.859$ at β^*, against a finite-sample ceiling of 0.944 (train vs. test) and -0.005 for the independent-site model, which matches f_i equally well. The regression slope shows that $\beta = 1$ under-estimates the correlation amplitudes (0.78), and β^* corrects most of this (0.85, against a ceiling of 0.88). Note that r and the slope measure different things and are related by slope $= r\,\sigma_{\text{gen}}/\sigma_{\text{nat}}$: r is the tightness of the relation and is blind to a uniform rescaling, the slope is the amplitude. Neither should be compared to 1 naively — the natural C_{ij} carry sampling noise, which attenuates any least-squares slope. A half-split of the test set puts their reliability at 0.94, and dividing by it gives disattenuated slopes 0.83 ($\beta = 1$), 0.91 (β^*) and 0.94 (ceiling). The low Spearman values on C_{ij} are expected: most of the 861 273 entries are sampling noise around zero, so their ranks carry little information; on the strongest 1% of the entries the rank agreement is high as well (Table 5).

Diversity and novelty (Fig. 7, Table 6).

- **Generated vs. test.** The closest test sequence lies on average at 0.44 ($\beta = 1$) and 0.41 (β^*). Two natural references bound this: a test sequence's closest *other* test sequence sits at 0.12 and its closest *training* sequence at 0.18. Natural sequences therefore have much closer natural neighbours than the generated ones do — the samples are new sequences, not near-copies.
- **Generated vs. train.** There is no memorisation: the mean distance to the closest training sequence is 0.42 (5% quantile 0.32) and every sample is distinct (100.0% unique).
- **Generated vs. generated.** The all-pairs distribution (mean 0.68 at $\beta = 1$, 0.65 at β^*) sits on top of the test–test distribution (0.60), without the left tail of near-identical natural pairs (phylogenetic redundancy removed by the weights). The samples are therefore neither collapsed onto a few modes nor spread too thin.

cluster	weight share		$\langle E \rangle$ natural		$\langle \log \mathrm{PL} \rangle$ natural		ΔF
	natural	generated	test	train	test	train	(nats)
C0†	0.141	0.051	-119.5	-122.6	-80.5	-71.8	+1.01
C1	0.387	0.352	-134.0	-134.5	-76.8	-71.1	+0.09
C2	0.473	0.597	-133.5	-137.2	-79.7	-71.0	-0.23

Table 3: Per-cluster diagnosis of the sampling weights ($\beta = 1$). Weight shares and $\langle \cdot \rangle$ use the reweighting weights for the natural sets. $\Delta F = -\ln(s^{\mathrm{gen}}/s^{\mathrm{nat}})$ is the model's free-energy error for the cluster, in nats. †: the cluster the sampler misses. The conditionals are fitted equally well everywhere ($\langle \log \mathrm{PL} \rangle$ on train), but the energy of C0 is far higher, so its Boltzmann weight is far too small.

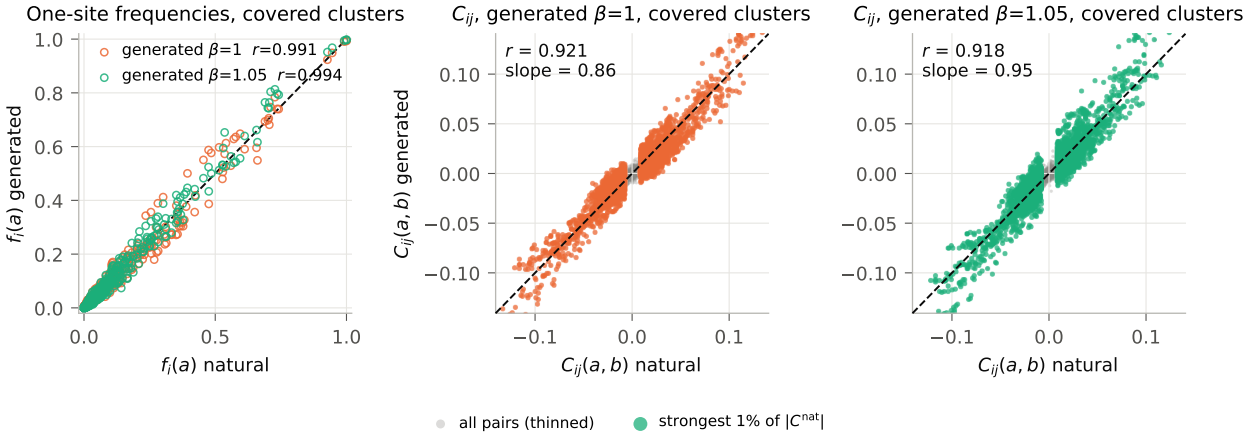


Figure 5: As Fig. 6, with natural and generated sets both restricted to the covered PCA clusters.

Energy of the samples (Fig. 8). The reweighted mean energy of the test sequences is -131.7 (unweighted -146.2). Samples at $\beta = 1$ have mean -125.9 (Wasserstein distance to test 5.9); at β^* the distribution lies on top of the natural one (-133.8, distance 2.1, independent-site model 39.5). Per-site pseudo-log-likelihoods show the same ordering: independent-site $\ll \beta = 1 < \beta^* \lesssim$ natural.

7 Effect of the split

Everything above uses the sequence-disjoint split of §3. Running the identical pipeline on the supplied species split (14 502 train / 3 739 test, 79% of the test set copied from the training set) gives the numbers in the second column below. The two differ in the direction one expects: the shipped split makes the model look better on the held-out likelihood, because most of its "test" sequences were seen during training.

	sequence-disjoint (this work)	supplied species split
train / test sequences	29 312 / 7 401	14 502 / 3 739
test sequences copied from train	0%	79%
validation $-\log \mathrm{PL}$ per sequence	82.18	85.61
test $-\log \mathrm{PL}$ per sequence	78.72	75.70
on its non-copied test sequences	—	77.13
contact PPV at $L/2$	1.00	0.97
$r(f_i)$, $\beta = 1$	0.990	0.990
$r(C_{ij})$, $\beta = 1$	0.871	0.865
ceiling (train vs. test)	0.944	0.960
restricted to covered clusters	0.918	0.921
generated $\to$ nearest train, $\beta = 1$	0.42	0.42

The generative metrics themselves are barely affected — the model learns nearly the same distribution either way — but the *interpretation* of the likelihood numbers changes completely, and only the disjoint split supports a claim of generalisation.

statistic	set vs. natural (covered clusters)	Pearson r	Spearman ρ	slope	RMSE
$f_i(a)$	generated, $\beta = 1$	0.991	0.970	0.98	0.0153
	generated, $\beta^* = 1.05$	0.996	0.969	1.03	0.0118
	natural train (ceiling)	0.998	0.980	0.98	0.0079
	independent-site model	0.993	0.973	0.95	0.0144
$C_{ij}(a, b)$	generated, $\beta = 1$	0.918	0.179	0.82	0.0012
	generated, $\beta^* = 1.05$	0.918	0.199	0.91	0.0013
	natural train (ceiling)	0.941	0.345	0.87	0.0010
	independent-site model	-0.013	0.060	-0.00	0.0031
C_{ij}, top 1%	generated, $\beta = 1$	0.961	0.919	0.87	0.0084
	generated, $\beta^* = 1.05$	0.961	0.919	0.97	0.0083
	natural train (ceiling)	0.979	0.938	0.91	0.0061
	independent-site model	-0.031	-0.035	-0.00	0.0296

Table 4: Statistics restricted to the covered PCA clusters (85.9% of the natural test weight). Compare with Table 5, which uses the full sets.

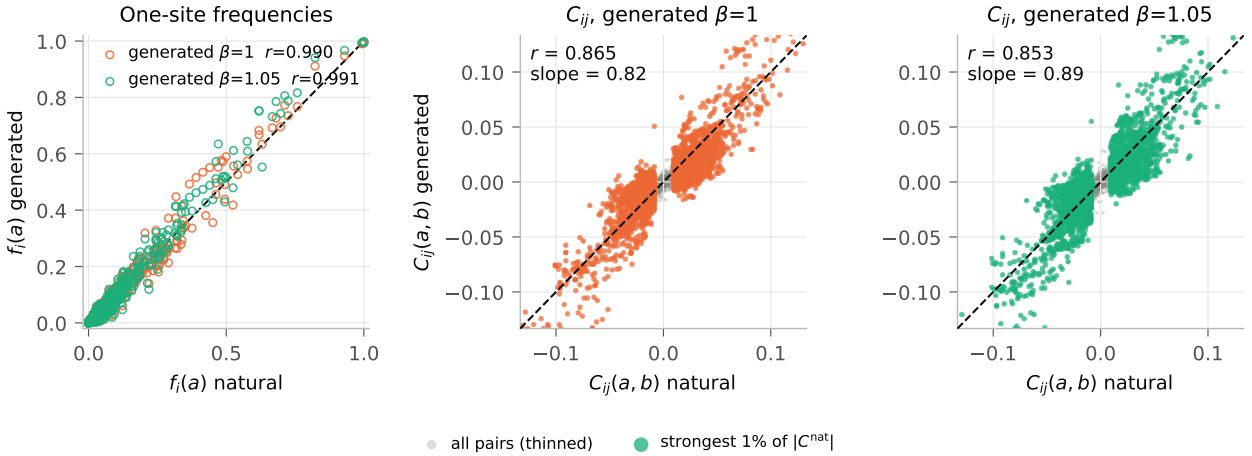


Figure 6: Generated vs. natural (reweighted test) one-site frequencies (1 323 values) and connected correlations. The C_{ij} panels show the strongest 1% of $|C^{\mathrm{nat}}_{ij}(a, b)|$ in colour over a thinned grey sample of all 861 273 entries. Dashed: identity.

8 Summary and caveats

- **Data.** The supplied split is species-disjoint but not sequence-disjoint (79% verbatim copies), so it cannot measure generalisation. The split used here clusters the full pool at 80% identity and moves whole clusters to one side (29 312 / 7 401 sequences, no exact copies, median distance 0.175 to the nearest training sequence).
- **Model.** The attention-factorised Potts model ($H = 128$, $d = 10$) predicts contacts well (PPV 1.00 at $L/2$) and keeps a 11% pseudo-likelihood gap to genuinely unseen sequences.
- **Sampling.** The vectorised sampler is 10^3–$10^5\times$ faster than the original code and decorrelates from its seed in about 11 sweeps ($\tau_{\mathrm{int}} = 6.5$). Samples are end-points of independent chains after 500 sweeps ($\approx 40\,\tau_{\mathrm{int}}$).
- **Generation quality.** $r(f_i) = 0.990$ and $r(C_{ij}) = 0.871$–0.859 against the 0.944 ceiling, rising to 0.918–0.918 (ceiling 0.941) on the modes the sampler populates. The samples are novel and as diverse as natural sequences. $\beta^* = 1.05$ restores the correlation amplitude and the natural energy level at a small cost in diversity.
- **Limitations.** (i) PL training does not constrain the global normalisation, hence the β offset. (ii) The attention couplings are asymmetric, so the PL chain is only approximately Boltzmann. (iii) The HscB-like sub-family is under-sampled (5.1% vs. 14.1%) because pseudo-likelihood fits conditionals and not the relative masses of the basins: the model learns that sub-family's local statistics as well as any other ($\langle \log \mathrm{PL} \rangle$ on train -71.8 vs. -71.1/-71.0) but puts its energy 14 units too high, a free-energy error of +1.01 nats. Adding explicit fields, fitting a mixture, or reweighting the training set towards the sub-family

statistic	set vs. natural test	Pearson r	Spearman ρ	slope	RMSE
$f_i(a)$	generated, $\beta = 1$	0.990	0.972	1.01	0.0160
	generated, $\beta^* = 1.05$	0.992	0.970	1.06	0.0165
	natural train (ceiling)	0.998	0.983	0.99	0.0070
	independent-site model	0.998	0.982	0.99	0.0072
$C_{ij}(a, b)$	generated, $\beta = 1$	0.871	0.193	0.78	0.0015
	generated, $\beta^* = 1.05$	0.859	0.209	0.85	0.0016
	natural train (ceiling)	0.944	0.377	0.88	0.0010
	independent-site model	-0.005	0.058	-0.00	0.0030
C_{ij}, top 1%	generated, $\beta = 1$	0.918	0.880	0.84	0.0113
	generated, $\beta^* = 1.05$	0.908	0.865	0.93	0.0123
	natural train (ceiling)	0.981	0.949	0.91	0.0057
	independent-site model	-0.009	-0.022	-0.00	0.0283

Table 5: Agreement of generated statistics with the natural test set (natural sets reweighted, generated sets uniform). "Top 1%": the 1% largest $|C_{ij}^{\text{test}}(a, b)|$.

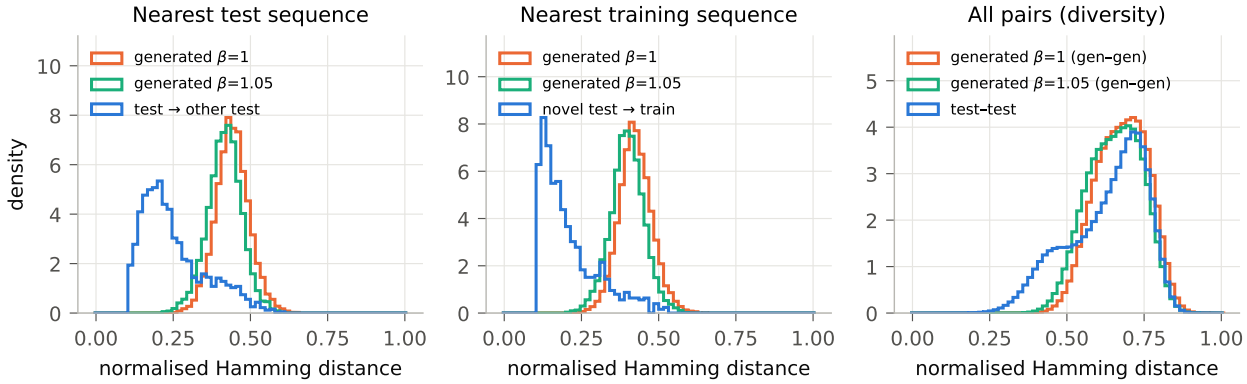


Figure 7: Normalised Hamming distances; triangles mark means. Each panel carries a different blue natural reference, and each answers a different question. **(a)** nearest-neighbour distance *into the test set*: for the generated sets, how far each sample is from the closest natural test sequence; the blue curve is the same quantity for a test sequence against the *other* test sequences (leave-one-out), i.e. how close natural relatives normally are. Its spike near zero is the family's phylogenetic redundancy. **(b)** nearest-neighbour distance *into the training set* — the memorisation check. Blue is the held-out test set measured the same way, which is the strongest "legitimate" proximity to training data a genuine J-domain shows. **(c)** all pairwise distances *inside* one set (3000 sequences per set): the spread of the set itself. Blue is the natural test set; the dotted grey curve is generated-vs-test, which lies on top of the generated curve when the two sets occupy the same region of sequence space.

would all address the prior directly. (iv) The split removes only exact copies, so 61% of the test set still lies within 0.2 of a training sequence; the held-out likelihood is a fair measure of generalisation to unseen sequences, not to unseen sub-families.

Code: `attdca/` (package), `scripts/` (pipeline), `tests/` (checks of the vectorised likelihood, energy and sampler against naive loops and an exact two-site Gibbs target).

distance (normalised by L)	mean	median	5% quantile	$= 0$
gen ($\beta = 1$) $\to$ nearest test	0.437	0.444	0.333	0.0%
gen ($\beta = 1$) $\to$ nearest train	0.420	0.413	0.317	0.0%
gen ($\beta = 1$) $\to$ nearest other gen	0.430	0.429	0.349	0.0%
gen–gen, all pairs ($\beta = 1$)	0.679	0.683	0.540	0.0%
gen–test, all pairs ($\beta = 1$)	0.666	0.683	0.492	0.0%
gen ($\beta^* = 1.05$) $\to$ nearest test	0.405	0.413	0.302	0.0%
gen ($\beta^* = 1.05$) $\to$ nearest train	0.389	0.397	0.286	0.0%
gen ($\beta^* = 1.05$) $\to$ nearest other gen	0.394	0.397	0.317	0.0%
gen–gen, all pairs ($\beta^* = 1.05$)	0.650	0.651	0.508	0.0%
gen–test, all pairs ($\beta^* = 1.05$)	0.639	0.651	0.460	0.0%
test $\to$ nearest other test	0.116	0.079	0.016	0.0%
test $\to$ nearest train	0.179	0.175	0.032	0.0%
novel test $\to$ nearest train	0.179	0.175	0.032	0.0%
test–test, all pairs	0.598	0.635	0.365	0.0%

Table 6: Hamming-distance summary; "= 0" is the fraction of exact copies.

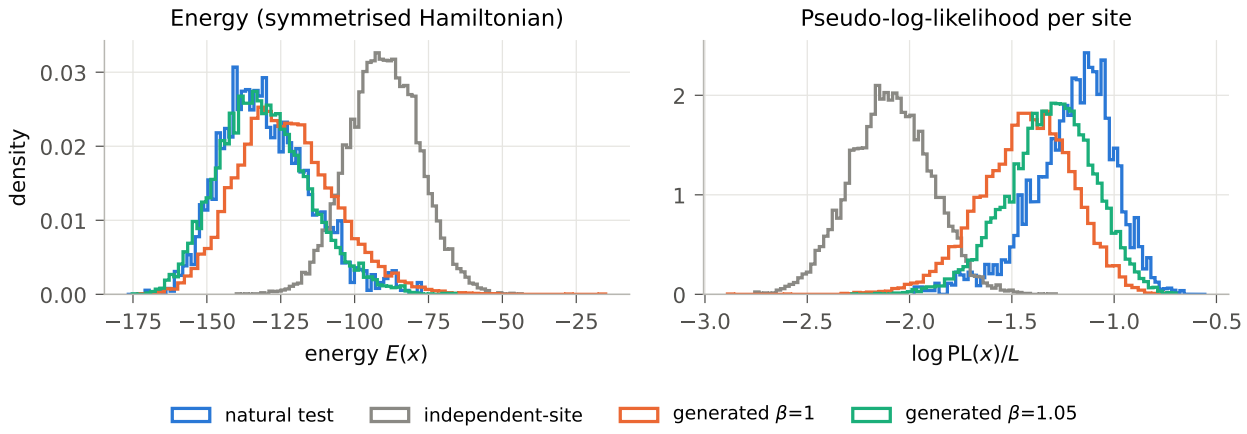


Figure 8: Distribution of the energy $E(\boldsymbol{x})$ and of $\log \mathrm{PL}(\boldsymbol{x})/L$ under the trained model.